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BreedAI: An open-source framework for fair and standardized machine learning-based genomic prediction in livestock

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Background

Genomic prediction is central to modern livestock breeding, yet existing workflows are difficult to compare, reproduce, and deploy. Production pipelines often rely on default GBLUP without systematic benchmarking, while research uses inconsistent preprocessing and validation.

BreedAI provides a unified, fair, and reproducible framework to standardize machine-learning evaluation and enable reliable deployment across livestock populations.

The BreedAI Framework

Transparency · Fair standardization · Reproducible pipeline.

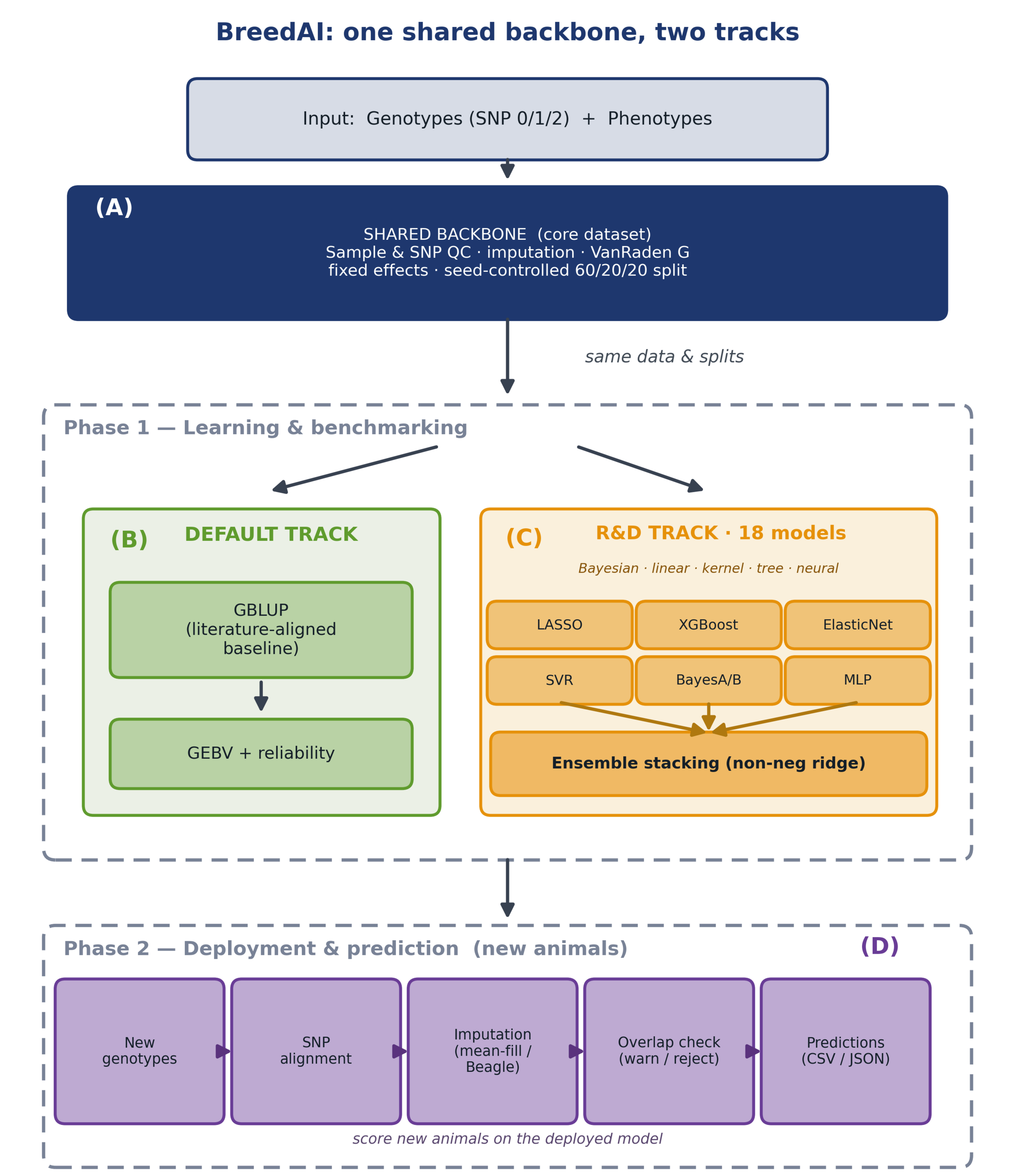
(A) Core dataset: one shared backbone for every method: sample & SNP QC · imputation · VanRaden G-matrix · fixed-effects modeling · seed-controlled 60/20/20 split (optional pedigree → H-matrix / ssGBLUP).

(B) Default track: literature-aligned GBLUP baseline → GEBV + reliability.

(C) R&D track: 18 models (Bayesian · linear · kernel · tree · neural) + stacking ensembles, on the same data & splits.

(D) Deployment (new animals): SNP alignment · imputation · overlap guardrails → predictions.

Scope: sequencing (FASTQ→VCF) is performed upstream / out of scope; BreedAI starts at genotypes.



Results: Benchmarking Public Dataset

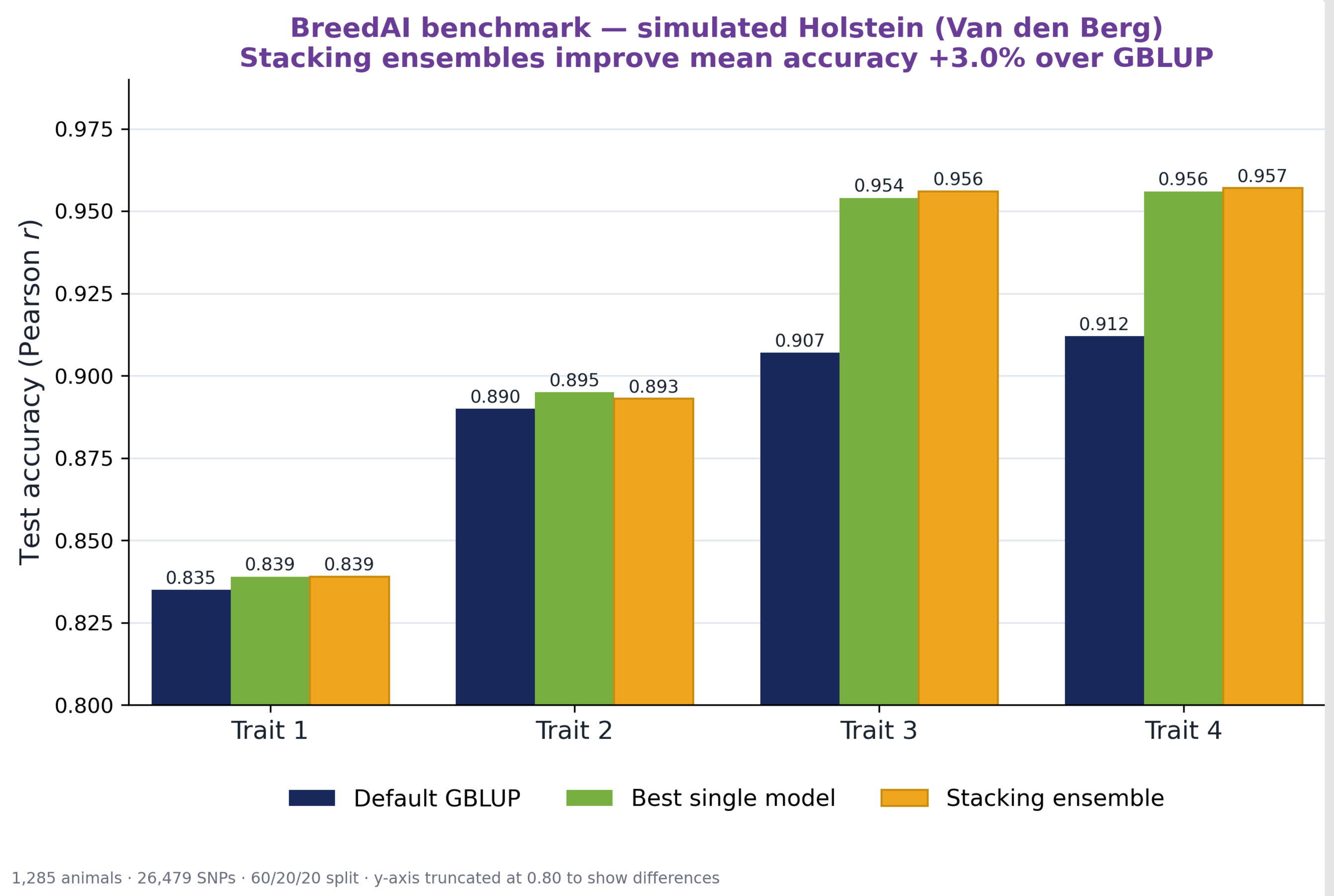
- **Fair benchmarking on a shared backbone:** all models trained and tested on the same preprocessed data and the same 60/20/20 split (seed-controlled).
- **Dataset:** public Van den Berg simulated Holstein cattle (QTL300, r_g = 0.8). After QC: 1,285 animals · 26,479 SNPs · 4 traits; split 771/257/257.
- **Models:** GBLUP baseline (RidgeCV) vs 18 algorithms across 6 families + 4 ensembles; stacking = non-negative ridge on out-of-fold predictions (nested CV, no label leakage).
- **Fixed effects:** intercept-only in this proof of concept.
- **Reporting:** R^2, Pearson r, RMSE, MAE, bias per model, machine-readable (CSV/JSON).

Cattle benchmark: penalized linear models and stacking outperform the default GBLUP on moderately polygenic traits.

Category	Mean test Pearson r	Range
Default GBLUP (RidgeCV)	0.886	0.835–0.912
Best single model (penalized linear)	0.911	up to 0.956
Best ensemble (stacking)	0.912	up to 0.957
Gain over GBLUP	3.00%	–

- The GBLUP baseline is genuinely strong (mean r = 0.886), validating the literature-aligned default.
- Penalized linear models beat it on moderately polygenic traits (up to r = 0.956).
- Non-negative ridge stacking reaches the highest accuracy (r = 0.957; +3.0% mean).

Tree-based and neural methods underperformed on this dataset, showing how fair comparison can reveal when complex ML does not improve prediction.



Why It Matters ?

What's new: a single shared backbone turns ad-hoc genomic prediction into an auditable, comparable, deployable workflow.

Fair: identical data, QC, and splits remove preprocessing/split confounds.

Reproducible: seed- and config-controlled, open-source; runs via menu or HPC scripts.

Deployable: Phase-2 prediction for new genotypes with SNP-overlap guardrails (warn <80%, reject <50%).

18	0.957	+3.0%	60/20/20
ML models benchmarked	best test accuracy (r)	stacking vs GBLUP	seed-controlled split

One shared backbone, two tracks → fair, reproducible benchmarking. Stacking ensembles add +3.0% over GBLUP.

Conclusion

BreedAI delivers a standardized, reproducible framework for machine-learning benchmarking and deployment in livestock genomic prediction, enabling fair comparison across models, traits, and species.

By holding preprocessing and splits constant across all models, it makes accuracy differences interpretable and results auditable. The framework is designed to scale across different species and to support genomic selection in under-represented and locally important populations.

Future Work and Availability

- Extends to more species (sheep, goats; under-represented populations such as camels and horses) and to multi-trait prediction.
- Code: github.com/nfdp-genome/BreedAI-Framework

Scan me!

- Built with Python (NumPy/pandas/scikit-learn) and R (BGLR), SLURM/Ibex.

Acknowledgment

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